NexPlast – Circular Economy Platform Documentation

Platform Vision

Plastic Waste → NexPlast Collection → Recycling Industries → Industries Pay NexPlast → Users Earn Coins → Coins Redeemed for Eco-friendly Products.

Business Model

Paying Customers: Plastic recycling factories, plastic manufacturers, FMCG companies, CSR partners, government bodies, and brands selling recycled products.

Revenue Streams
1. Selling collected plastic.
2. Marketplace commission.
3. Brand partnerships (EPR).
4. Carbon credits.
5. Government contracts.

Example Economics

20 kg PET bottles collected. Factory pays ■32/kg = ■640 revenue. Costs: Pickup ■120 + Sorting ■60 + Rewards ■100 = ■280. Estimated Profit: ■360.

Coin System

Nature Coins represent environmental contribution, not cash. Rewards depend on weight, quality, and plastic type.

Marketplace

Examples: Recycled Bottle – 20 Coins, Flower Pot – 70 Coins, Chair – 450 Coins, Dustbin – 650 Coins.

Hybrid Payment Model

Instead of a coin-only economy, products should support flexible payments.

Example:
Recycled Flower Pot
• ■199
OR
• 80 Nature Coins
OR
• ■99 + 40 Nature Coins

This allows new users with zero coins to purchase products while still encouraging recycling.

User Roles

Customer, Collector, Recycler, and Admin.

Workflow

User submits plastic → AI verifies → Pending Review → Collector accepts → Pickup scheduled → Collector verifies quantity and weight → Uploads proof → User signs digitally → Pickup completed → Coins credited.

AI Verification

YOLOv8 detects bottles, bags, containers, wrappers, cups, buckets, caps, etc. Non-plastic is rejected, counts are compared with user input, and low-confidence results are sent for manual review.

Bulk Submission Rule

Users do not need one image per item.

Example: 100 bottles → Spread bottles → Take ONE photo → AI counts automatically.
Plastic bags → One photo.

Rule: One category = One image.

Collector Verification

Collector updates actual quantity and weight, uploads proof photo, and the system recalculates rewards automatically.

Wallet & Gamification

Nature Wallet tracks earned, spent, and pending coins. Bronze, Silver, Gold, and Platinum levels, badges, achievements, leaderboards, and monthly challenges.

Environmental Dashboard

Displays plastic recycled, trees saved, CO■ reduced, ocean plastic prevented, and user ranking.

Updated Nature Coin System (Replaces Previous Coin Section)

Nature Coin Policy
- Nature Coins represent environmental contribution, not cash.
- Base Rate: 1 Nature Coin = 100 g of standard recyclable plastic.
- 1 g = 0.01 Coin
- 100 g = 1 Coin
- 1 kg = 10 Coins
- 10 kg = 100 Coins
- Plastic type and quality may increase or decrease coin rewards.

Internal Reward Value
- 1 Nature Coin has an approximate internal reward value of Rs0.50.
- Coins cannot be exchanged for cash.
- Coins can only be redeemed in the NexPlast Marketplace.

Updated Business Example
User recycles 20 kg of clean PET bottles.
Plastic collected = 20,000 g = 200 Nature Coins.
Factory Purchase Price: 20 kg x Rs32/kg = Rs640 Revenue.
Operational Costs:
  Pickup = Rs120
  Sorting = Rs60
  Reward Budget = Rs100 (200 Coins x Rs0.50 internal value)
Total Cost = Rs280
Estimated Profit = Rs360

Marketplace Examples
- Eco Bottle: 40 Coins
- Flower Pot: 100 Coins
- Dustbin: 300 Coins
- Chair: 600 Coins

Note:
The Rs32/kg factory price is an example only and varies by plastic type and market conditions. Coins are an incentive system, not cash.